

Causal Foundations and Experimental Certification of Contingent Assignment, Heredity, and Reflexive Maintenance

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Abstract

A reproducible input–output relation alone need not reveal whether its mapping was present at the origin or formed later. We formulate causal criteria for contingent operational assignment, assignment-specific causal heredity, and one-step maintenance of future assignment capacity. Building on an earlier construction in a specified reaction–compartment model, we study these questions without treating that architecture as universal. Assignment-neutral dynamics admit contingent realized mappings, while symmetry-fixed and quotient-erased states obey distinct impossibility results. Passive observation and interventions on inputs alone do not generally identify decoder-free provenance. Under specified causal assumptions, interventions on organization, lineage, and support yield separate certificates. We give exact conditions for combining them in one system, a certificate for actual future assignment capacity, and a finite construction showing that the resulting hypothesis class is nonempty. Finite-error and finite-sample guarantees retain the hard causal requirements. Resource feasibility is characterized by causal residual viability in a commutative monoid; minimality is relative to a specified premise language. The Empirical Correspondence System translates the mathematical conditions into physical requirements, interventions, observables, controls, and prespecified experimental decisions. It distinguishes whether a system meets the assumptions, whether the assessment is adequate, and what the empirical evidence supports. Global component calibration is separated from confirmatory mapping outcomes. The selected literature supplies component precedents but does not establish a full joint realization here. A worked RNA-routing design remains conditional on unresolved calibration and addressability. No new laboratory data, chemical realization, historical origin, or priority claim is reported.

Keywords: contingent assignment; causal heredity; reflexive maintenance; causal intervention; identifiability; experimental certification; residual viability.

1. Introduction

A reproducible input-output relation need not be a contingent assignment. A fixed physical transformation can be informative without its mapping depending on an endogenous organization. Parent-descendant resemblance can arise from common forcing. An output can correlate with persistence without causally maintaining future assignment capacity. These distinctions matter both when constructing an emergence model and when deciding what an experiment has established.

The random-matching model in Section 3.4 gives a concrete example. Consider two inputs and two outputs, with no links initially. Unmatched input-output pairs form links at equal rates. During the assignment assay, link formation is clamped in the same way for every challenge, and the response identifies the current partner of the challenged input, or a common null symbol if it has no partner. Removing a realized link changes its challenged response. The two possible complete matchings give different input-output relations under the same linking rule. Here the rule is specified, while the matching forms during the process. The linking and assay rules are assumptions of this construction; their molecular origin is not established here.

The direct predecessor is the model-specific construction in [28]. That study used one fixed, independently assignment-symmetric reaction–compartment grammar. It proved that contingent readout, topology-dependent causal heredity, and output-mediated maintenance coexist in a common admissible parameter regime. The construction uses neither a mapping-quality reward nor an externally activated reaction stage. It studied assignment formation within a specified recognition, assembly, templating, and readout architecture. It did not derive the molecular origin of the supplied recognition, assembly, copying, and readout capabilities. The present article does not treat the earlier species, reaction grammar, parameterization, or compartment mechanism as universal premises.

We specify causal model classes and study assignment, its genesis, causal heredity, reflexive maintenance, robustness, and resource feasibility. Each property has its own hypotheses. Heredity and reflexive maintenance are distinct, and combining local results in one system requires compatible boundary conditions, interventions, and causal ordering. We also establish identifiability limits and finite-error guarantees, and

specify what an experiment must assess. An explicit finite construction shows that the joint conditions can be satisfied; it does not select a chemistry.

The autonomous physical law may not use a target mapping, translation-correctness score, mutual-information objective, semantic-success reward, modern-code similarity, or future-use variable. Information quantities and capacity sets are used only as diagnostics. These restrictions apply to the physical dynamics, including the specified link-formation model above.

For an experimental application, the question is whether a physical system and its interventions satisfy the hypotheses of a particular result. The Empirical Correspondence System (ECS) translates those hypotheses into assessable physical requirements and prospective experimental decisions. Mathematical definitions and proofs determine the conditions being tested. An empirical match does not prove the theorem. A failed physical correspondence does not refute it unless a mathematical defect is separately established.

Each result has a stated scope: pointwise, deterministic, finite-horizon, almost-sure, expected, metastable, stationary, local-linear, or asymptotic. Its conclusions are restricted to that scope. The article makes no chemical, prebiotic, historical, or priority claim.

Sections 2–13 give the mathematical definitions, results, counterexamples, and scope distinctions. Section 14 explains their dependencies. Sections 15–16 develop the experimental correspondence and validation methods. Section 17 places the results in context and states their limits. The Mathematical Supplement provides the full supporting constructions and condition dictionaries. The Experimental Methods Supplement gives the detailed evidence discussion, calibration protocol, and conditional RNA-routing design.

2. Causal-interventional substrate and operational assignment

2.0 Notation and certificate names

Under their respective assumptions, organization interventions test how the response changes, parent and lineage interventions test causal transmission to descendants, and support interventions test effects on later states. The maintenance results distinguish later support response from actual future assignment capacity. Composition asks whether the local results can hold in one causal system. The error analysis bounds quantitative deviations while retaining the structural causal requirements.

The abbreviations used below include OA (operational assignment), SCM (structural causal model), TV (total variation), and IID (independent and identically distributed).

Symbols are local to each stated model. In particular, the finite challenge set (A) in the assignment assay is distinct from the organization random variable (A) in the later causal certificates. Each role is reintroduced where used. DFI-A and DFI-H denote the organization and heredity certificates; DFI-R(loop) denotes the support-loop/re-entry certificate, whereas DFI-R(cap) denotes the actual-future-capacity certificate. UCCE and UFE denote the exact composition and finite-error certificates, respectively. These are names for the printed mathematical conditions, not additional assumptions or levels of empirical confirmation.

Domain convention. The substrate may be measurable, but a formula written with point probabilities and sums uses a finite or countable discrete response/context alphabet; its challenge set is finite whenever a maximum over challenges is used. The finite-alphabet inference guarantees require finite response alphabets. No extension of those point-mass formulas to arbitrary nonatomic response spaces is asserted. Measure-theoretic statements explicitly written with kernels, integrals, or total variation retain their own stated domains. The finite acyclic construction and all finite certification examples below satisfy these conventions.

Measurable-event scope. Every set to which a probability or kernel is applied below is required to belong to its declared sigma-field. In particular, operational-assignment and capacity sets used probabilistically, their indicators, signature and state-binding events, conditioning cells, and the preimages under the maps used in those expressions must be measurable. Structural response laws must be measurable probability kernels; any conditional kernels used in a measurable extension must exist on the declared spaces. Signature maps and probability-transporting isomorphisms are measurable, with measurable inverses for the latter. These are explicit domain hypotheses, not consequences of the word “event” or of individual intervention measurability. Mathematical Supplement S19 proves sufficient conditions for fixed countable diagnostic families and gives the downstream probability dictionary. Finite discrete constructions satisfy these conditions directly. An uncountable existential diagnostic or signature-equality condition receives no automatic measurability claim; its required measurable sets and kernels must be established separately.

before applying a probabilistic conclusion.

2.1 Primitive substrate

An admissible causal-interventional system is represented by

$$\mathcal{M} = (\mathcal{S}, \mathcal{K}, \mathcal{J}),$$

where:

- $\mathcal{S}$ is a typed local measurable causal-state/factorization structure;
- $\mathcal{K}$ is the adapted structural-law family, including root, preparation, exogenous, boundary, and endogenous kernels;
- $\mathcal{J}$ is an admissible intervention family with composition semantics.

Autonomous kernels are adapted to the declared causal filtration. No autonomous transition can read future state, future success, or future usefulness.

Lineage, causal heredity, reflexive support, and resource bookkeeping are derived structures on this substrate rather than additional primitive semantic variables.

2.2 Future-interventional state

At a causal cut τ , let $\mathcal{W}_\tau$ be a separating/generating family of future intervention words. For a physical state x , define its future-interventional signature

$$\Sigma_\tau(x) = (P_x^w)_{w \in \mathcal{W}_\tau}.$$

Two states are equivalent relative to the declared future intervention class if their signatures agree. Any statistic sufficient for all declared future intervention laws must refine this signature equivalence.

This representation prevents a notation change from changing the scientific property: states are identified by future causal behavior, not by analyst labels.

Equivalence defined by future input–output behaviour has a precedent in the causal states of input–output processes and the ε -transducer [33]. Here the signature is indexed by the explicitly declared future intervention words at a causal cut. We do not identify this construction with a stationary input–output process or import the optimality results of that framework without their assumptions.

2.3 Assignment assay, symmetry action, and assay compatibility

An assignment assay consists of:

- a finite challenge set A ;
- a response set B ;
- challenge interventions $\{J_a\}_{a \in A}$;
- a response readout Y ;
- a nontrivial group Γ .

The group acts on physical causal states, challenges, and responses. Write these actions as

$$c \mapsto \gamma c, \quad a \mapsto \gamma_A a, \quad b \mapsto \gamma_B b.$$

For a causal state c , define the challenged response channel

$$Q_c(b \mid a) = P_c^{J_a}(Y = b).$$

The induced action on channels is

$$(\gamma \cdot Q)(b \mid a) = Q(\gamma_B^{-1} b \mid \gamma_A^{-1} a).$$

The assignment assay is equivariant when

$$Q_{\gamma c} = \gamma \cdot Q_c.$$

The full law/preparation/boundary/intervention system is **assignment-neutral** when the declared Γ -action is an automorphism of those physical structures and no target map, mutual-information objective, semantic-success variable, or future-use variable enters the path law.

A state supports **operational assignment** when:

1. at least one challenge contrast is nonzero; and
2. there exists an admissible intervention on the pre-challenge endogenous organization that changes

the challenge-response channel under the declared matched comparison.

The assignment is **contingent** when symmetry-related endogenous organizations under the same law realize distinct channels in a nontrivial Γ -orbit.

An **assay-compatible causal-interventional isomorphism** between two formulations is a bijective transport of causal states, structural kernels, admissible interventions, challenges, responses, readouts, and symmetry actions that preserves every challenged interventional law. Operational assignment and contingency are invariant under such an isomorphism.

2.4 Decoder-free origin and diagnostically complete origin audit

Decoder freedom is relative to an explicitly declared origin ontology and relation. It concerns the absence of an already stored, concrete decoder-bearing organization, including a dormant one. Symmetry of its preparation law alone does not establish this absence.

Fix a family $\mathcal{D}_0$ of origin diagnostic protocols. A protocol may expose an existing organization by removing an inhibitory gate, challenge it, and delete or neutralize it. It may be a finite intervention word. During the diagnostic word, formation of a new decoder-bearing relation is disabled. A delayed read is admissible only when it preserves this origin-versus-new-formation distinction. The family is fixed before outcomes, Γ -stable, nonconstructive, and independent of target mapping, semantic success and future use.

For an actual origin state x , let Q_x^d be the challenge-response channel after protocol d , and put

$$\mathcal{R}_0(x) = \{Q_x^d : d \in \mathcal{D}_0\}.$$

The state x includes any realized origin organization; channels are not first averaged over a random preparation of different stored mappings. Write $\text{Pre}_\mathcal{O}(x)$ for the independently specified structural property that a concrete decoder-bearing organization is present in the declared ontology. It is not defined to mean a positive audit result.

The diagnostic-completeness condition is

$$\text{DC0} : \quad \text{Pre}_\mathcal{O}(x) \implies \text{some } Q_x^d \text{ exposes an organization-dependent} \\ \text{challenged relation with a nontrivial } \Gamma\text{-orbit.}$$

The quantifier ranges over the declared admissible origin states. Thus every potentially relevant stored relation must be detectable through the declared repertoire; the ability to delete it without any detectable response does not suffice. DC0 needs a structural proof or a separately justified membership assessment. A collection of null readings does not establish DC0.

The full decoder-free origin criterion requires assignment-neutral origin laws, preparations, boundaries and interventions, DC0, and absence of the specified nontrivial relation in $\mathcal{R}_0(x)$ for every allowed realized origin state (or almost surely for an explicitly probabilistic claim). By contraposition of DC0, the last two conditions exclude $\text{Pre}_\mathcal{O}(x)$. No absence beyond the declared ontology is inferred.

A potentially relevant organization that is inaccessible or can only be deleted without exposure leaves the audit **inconclusive**. In the empty-matching construction of Section 3.4, structural absence is proved directly: there are no links at origin and no diagnostic word creates one. The generic readout law is not a concrete preinstalled matching.

This origin criterion is distinct from the later DFI-A certificate. DFI-A uses its explicit null organization and exhaustive M1 ontology to certify a relation-relative origin statement plus a later organization-dependent response. It does not alone certify full origin-law neutrality or spontaneous genesis.

Representation principle

Fix the scientific object: the declared origin ontology $\mathcal{O}$, the independent predicate $\text{Pre}_\mathcal{O}$, origin cut, admissible realized origin domain, assay, and diagnostic family. An origin-preserving isomorphism must transport all these data, as well as the causal laws, interventions, and symmetry action. In particular, if f transports origin states, require

$$\text{Pre}_{\mathcal{O}'}(f(x)) \iff \text{Pre}_\mathcal{O}(x), \quad Q_{f(x)}^{d'} = f_{B*} Q_x^d$$

with the transported challenge argument understood and d' the transported protocol. Under such an isomorphism, decoder-free-origin and operational-assignment verdicts are invariant.

Proof. Bijections carry the admissible origin states and diagnostic protocols onto their declared counterparts. Channel transport preserves equality and the nontriviality of symmetry orbits. Transport of the

structural predicate therefore carries both the premise and conclusion of DC0 in both directions. The null-response criterion, origin neutrality, and their contraposition are preserved; the same channel argument proves operational-assignment invariance. Changing the ontology, discarding part of the structural predicate, or moving the origin cut changes the scientific object and is outside this assertion. $\square$

3. Contingent assignment genesis

3.1 Theorem 1 — Fixed-set impossibility

Let

$$\text{Fix}(\Gamma) = \{c : \gamma c = c \ \forall \gamma \in \Gamma\}.$$

If the channel map $\Phi(c) = Q_c$ is equivariant, then for any $c \in \text{Fix}(\Gamma)$,

$$\Phi(c) = \Phi(\gamma c) = \gamma \cdot \Phi(c),$$

so the channel orbit is trivial. Therefore an exactly symmetry-fixed realized state cannot itself support contingent assignment.

If deterministic evolution is exactly equivariant,

$$F_t(\gamma c) = \gamma F_t(c),$$

then

$$F_t(\text{Fix}(\Gamma)) \subseteq \text{Fix}(\Gamma),$$

and an exactly fixed realized origin cannot escape to a contingent assignment state.

The stochastic statement here concerns an actual update sequence indexed by $k \in \mathbb{N}_0$, or a finite prefix. Let C_k be adapted to $(\mathcal{F}_k)$, with measurable fixed sets $D_k = \text{Fix}_{\mathcal{C}_k}(\Gamma)$. Let $K_k(\omega, \cdot)$ be an $\mathcal{F}_k$ -measurable probability kernel that is the actual conditional law of C_{k+1} given $\mathcal{F}_k$. Require $P(C_0 \in D_0) = 1$ and, for every update index k ,

$$\mathbf{1}_{\{C_k \in D_k\}} K_k(\omega, D_{k+1}) = \mathbf{1}_{\{C_k \in D_k\}} \quad \text{a.s.}$$

Then

$$P\{C_k \in D_k \text{ for every } k \in \mathbb{N}_0\} = 1.$$

Proof of the stochastic statement. Conditional expectation gives $P(C_k \in D_k, C_{k+1} \notin D_{k+1}) = 0$. Starting with the origin hypothesis, induction gives $P(C_k \notin D_k) = 0$ for each k ; the countable union of these null events has probability zero. Equivariance of the assay at each update then makes every such realized channel orbit trivial on this one probability-one event. $\square$

The kernels in this premise are the actual update laws, not merely transitions between every pair of deterministic observation times. The simultaneous conclusion ranges over the declared update sequence. No conclusion about unobserved intermediate values, accumulation-time limits outside that sequence, or arbitrary continuous-time sample paths is asserted here.

Claim class: pointwise deterministic impossibility; almost-sure simultaneous impossibility on the stated finite/countable actual update sequence.

3.2 Theorem 2 — Symmetry-quotient erasure

Let $\pi : \mathcal{C} \rightarrow \mathcal{C}/\Gamma$ be the orbit quotient. Let the reachable state domain $D \subseteq \mathcal{C}$ be Γ -stable, and let the challenged-channel map Φ be equivariant on D . If on D

$$\Phi = \bar{\Phi} \circ \pi,$$

then

$$\Phi(\gamma c) = \Phi(c)$$

while equivariance also gives

$$\Phi(\gamma c) = \gamma \cdot \Phi(c).$$

Hence every reachable channel is symmetry-fixed and contingent assignment is impossible.

The domain condition follows, for example, when the allowed preparation set is group-stable and the admissible transition/intervention family is equivariant. It is not inferred for an arbitrary conditioned

singleton realization. On the stated domain, assignment genesis requires output-relevant causal information that survives beyond the assignment-erasing quotient.

3.3 Theorem 3 — Equal-orbit first hitting

Let a finite group Γ act measurably on the state and path spaces, without changing time, with $X_t(\gamma\omega) = \gamma X_t(\omega)$, and transitively on a finite nonempty set Θ . Let the measurable orientation cells B_θ be pairwise disjoint and satisfy $\gamma B_\theta = B_{\gamma\theta}$. For a process X_t , with time domain $\mathbb{T} = \mathbb{N}_0$ or $[0, \infty)$, fix a finite horizon T and define

$$H_T = \{\omega : \exists t \in \mathbb{T} \cap [0, T], X_t(\omega) \in \bigcup_{\theta \in \Theta} B_\theta\}.$$

Assume H_T is measurable. Outside a Γ -invariant null set, every path in H_T must attain an actual first entry

$$\tau_T(\omega) = \min\{t \in \mathbb{T} \cap [0, T] : X_t(\omega) \in \bigcup_{\theta} B_\theta\}, \quad X_{\tau_T}(\omega) \in B_{\vartheta_T(\omega)}.$$

Assume the first-entry time and the unique orientation label ϑ_T are measurable, extending them by $+\infty$ and a fixed cemetery label off the attained-hit event. If the complete path law is Γ -invariant, then, for every $\theta \in \Theta$,

$$P(H_T \cap \{\vartheta_T = \theta\}) = \frac{P(H_T)}{|\Theta|}.$$

If, in addition, $P(H_T) > 0$, the conditional first-hit orientation is uniform:

$$P(\vartheta_T = \theta \mid H_T) = \frac{1}{|\Theta|}.$$

Proof. The action preserves time and transports the cells. Thus H_T is invariant and, on the attained-hit event, $\tau_T(\gamma\omega) = \tau_T(\omega)$ and $\vartheta_T(\gamma\omega) = \gamma\vartheta_T(\omega)$. Path-law invariance gives equal probabilities to the label events related by any γ . Transitivity makes their probabilities all equal. They are disjoint and exhaust H_T up to the declared null set, so their sum is $P(H_T)$. Division by $P(H_T)$ is used only when it is positive. ■

Actual entry is an assumption, not merely the infimum of visit times. Continuous sample paths alone do not ensure that an entry into arbitrary measurable cells is attained. Discrete-time first visits, and the finite matching jump process below, satisfy the required first-entry convention. This theorem is an ensemble-neutrality result; it does not assert that each realized path is symmetric.

3.4 Theorem 4 — Explicit decoder-free genesis

Let

$$U = \{u_1, \dots, u_n\}, \quad V = \{v_1, \dots, v_n\}, \quad n \geq 2.$$

A physical state is a partial matching $M \subseteq U \times V$, initially $M_0 = \emptyset$. Whenever u and v are unmatched, a link forms at the same rate $\lambda > 0$, independent of assay identity. Links persist through the genesis phase. More explicitly, this is the continuous-time Markov chain whose only genesis transitions are admissible one-link extensions, each with rate λ .

The symmetry group is

$$\Gamma = S_n(U) \times S_n(V).$$

During the assignment assay, link formation is clamped identically for every challenge. Challenge u returns its current matched partner v , if any, and a common null symbol otherwise. Single-realized-link ablations are admissible organization interventions.

If $k < n$ links exist, there are $(n - k)^2$ admissible pairs and total next-link rate

$$\lambda(n - k)^2 > 0.$$

Each jump increases matching size by one. The holding time at size k is exponential with parameter $\lambda(n - k)^2$; the sum of these n holding times is finite almost surely. The n -th jump is therefore an attained first entry into the perfect matchings, with

$$E[\tau_{\text{perfect}}] = \frac{1}{\lambda} \sum_{j=1}^n \frac{1}{j^2} < \infty.$$

Every perfect matching supports operational assignment because ablating a realized link changes its challenged response. Distinct perfect matchings are symmetry-related contingent organizations. Uniformity also follows by direct counting, independently of Theorem 3: at size k , each next pair has probability $1/(n-k)^2$. A specified perfect matching has $n!$ possible link orders; each order has probability

$$\prod_{j=1}^n \frac{1}{j^2} = \frac{1}{(n!)^2}.$$

The order events are disjoint, so

$$P(M_\infty = M) = n! \frac{1}{(n!)^2} = \frac{1}{n!}.$$

For every fixed $T > 0$, the hit event also has positive probability: requiring each of the n holding times to be at most T/n is a positive-probability sufficient event. Thus the finite-horizon conditional form of Theorem 3 is applicable as well.

The origin is decoder-free relative to the declared assay-complete nonconstructive origin audit because no origin link exists and no permitted origin minor can create a challenge-specific channel.

Scope: this theorem proves genesis only. It does not prove heredity, reflexive maintenance, robustness, resource admissibility, chemical realization, or historical occurrence.

3.5 Which genesis assumptions are necessary?

Theorem 1 excludes deterministic equivariant genesis from an exactly symmetry-fixed realized origin. This restricted impossibility does not, by itself, prove that stochastic transitions are necessary in every other origin class. Theorem 4 supplies a sufficient stochastic genesis construction. No alternative deterministic decoder-free genesis construction is established here.

Finite-time genesis and permanent retention of a named orientation are different targets. Theorem 4 concerns the former; the recurrent-erasure result in Section 8.1 limits the latter. Neither target is silently substituted for the other.

4. Identifiability and causal assignment certification

Property existence and property identifiability are distinct mathematical questions.

4.1 Theorem 5 — Passive observational no-go

There exist finite witness pairs with identical passive input-output path laws but different decoder-free provenance: in one model the assignment organization is created after the origin cut; in the other an assignment organization is already present but remains dormant until the same activation time.

Consequently, on any model class containing both witnesses,

$$\boxed{\text{passive I/O law alone} \not\Rightarrow \text{decoder-free provenance.}}$$

No passive statistic, arbitrarily large passive sample, or randomized decision rule can distinguish two models whose passive path measures are exactly equal.

The theorem is model-class relative: if a dormant-preinstalled alternative is excluded by assumption, passive data have not empirically distinguished it.

4.2 Theorem 6 — Input-intervention-only no-go

The no-go persists for arbitrary adaptive input policies when the allowed intervention interface acts only on declared inputs and cannot inspect, set, or condition on the hidden assignment-bearing organization.

Thus even active experimental control of all declared inputs does not universally identify decoder-free provenance when the provenance-bearing internal state remains causally inaccessible to the allowed protocol.

This result does **not** say intervention is useless. It identifies the boundary of the restricted intervention class.

4.3 Relation-relative internal-organization certificate

This certificate must not be confused with the full decoder-free-origin definition of Section 2.

Let the later internal organization state be A , with a unique null state $\perp$, nuisance/context state D , challenge/input U , output V , and certification protocol label J .

The structural boundary $M3^\dagger$ requires one stable output equation

$$V = F_V(U, A, D, E_V)$$

whose endogenous parents are exhausted by U, A, D , whose exogenous output noise is separated from the upstream protocol/state/context generation, and whose structural mechanism remains unchanged under the certification interventions.

The ontology boundary $M1$ requires that every **internal organization** capable of carrying a decoder-equivalent relation relevant to the certified $U \rightarrow V$ relation is represented by A . It does not claim that arbitrary structure in the fixed law itself is origin-neutral.

The certificate uses:

$$C0: \quad P(A_0 = \perp) = 1,$$

two later organization states a_0, a_1 and admissible joint surgical regimes

$$J_i(u) = do(A = a_i, U = u), \quad i \in \{0, 1\}.$$

There is no direct $J \rightarrow V$ or omitted protocol-to-output path. For each compared challenge u , $C2^*$ requires the same nuisance law $\mathcal{L}(D \mid J_i(u)) = \mu$ in both arms. The output noise has the same law λ_V and is independent of D under those regimes. The condition $C1^*$ requires positive separation of $\mathcal{L}(V \mid J_0(u))$ and $\mathcal{L}(V \mid J_1(u))$ for at least one common challenge u . If a different common law is used for different challenges, it must be indexed explicitly; here one fixed μ is used.

For a nuisance law μ , define

$$\mathcal{K}_{a,\mu}(v \mid u) = \sum_d P_{\lambda_V}[F_V(u, a, d, E_V) = v] \mu(d).$$

Under each joint regime, the stable equation and the specified product law of (D, E_V) give $\mathcal{L}(V \mid J_i(u)) = \mathcal{K}_{a_i,\mu}(\cdot \mid u)$ by the law of total probability. Thus the observed separation in $C1^*$ is the same structural-channel separation used in Δ_A . An observational conditional $V \mid U = u$, or a context law matched only before setting the challenge, cannot replace this premise.

Then

$$M1 + M3^\dagger + C0 + C1^* + C2^* \implies \text{later relation-relative } A\text{-dependent causal response.}$$

The quantitative separation is

$$\Delta_A = \max_u d_{TV}(\mathcal{K}_{a_0,\mu}(\cdot \mid u), \mathcal{K}_{a_1,\mu}(\cdot \mid u)) > 0.$$

Scope

The DFI-A theorem certifies:

- no preinstalled decoder-bearing **internal organization within M1** at the declared origin state;
- a later organization-dependent structural response relation.

It does **not** by itself certify:

- assignment-neutrality of the entire origin structural law;
- the full origin-minor/repertoire criterion;
- spontaneous or endogenous genesis;
- first onset.

The decoder-free genesis result requires both the origin-neutrality conditions in Section 2 and the genesis conditions in Section 3.

The J_i regimes are surgical experimental interventions that set the later endogenous organization; they do not demonstrate spontaneous formation. The structural condition $M3^\dagger$ excludes a direct effect of the protocol on the output that bypasses that organization.

Architecture-relative minimality

For the composite target consisting of the specified null origin together with the later organization-dependent response, $C0, C1^*, C2^*$ are deletion-indispensable within this certificate architecture and $M1, M3^\dagger$ are semantic/structural boundaries. This is not a global weakest-condition theorem. Explicit deletion/bypass countermodels are printed in the Mathematical Supplement.

5. Assignment-specific causal heredity

5.1 Heredity contrast and causal screening

Let $I_c, I_{c'}$ be physically realizable parent-organization interventions at the same causal cut under one matched boundary law. Let A_{n+1} be the descendant assignment-channel diagnostic and define

$$\nu_c = \mathcal{L}(A_{n+1} \mid I_c, \beta), \quad \nu_{c'} = \mathcal{L}(A_{n+1} \mid I_{c'}, \beta),$$

$$\delta_{\text{desc}} = \|\nu_c - \nu_{c'}\|_{\text{TV}}.$$

A state H is a causal screening cut state for the comparison when there exists a **common** downstream Markov kernel T such that

$$\nu_c = \mu_c T, \quad \nu_{c'} = \mu_{c'} T,$$

with

$$\mu_c = \mathcal{L}(H \mid I_c, \beta), \quad \mu_{c'} = \mathcal{L}(H \mid I_{c'}, \beta).$$

Merely intersecting every directed graph path is not sufficient; the cut must screen the compared regimes with one downstream kernel.

5.2 Theorem 7 — Transmission bottleneck

Let

$$\delta_H = \|\mu_c - \mu_{c'}\|_{\text{TV}}$$

and define the Dobrushin contraction coefficient of the downstream Markov kernel by

$$\eta(T) = \sup_{h, h'} d_{\text{TV}}(T(h, \cdot), T(h', \cdot)).$$

Then

$$\delta_{\text{desc}} \leq \eta(T) \delta_H \leq \delta_H.$$

Proof. The signed measure $\sigma = \mu_c - \mu_{c'}$ has total mass zero. For any descendant event B , $f_B(h) = T(h, B)$ has oscillation at most $\eta(T)$. Subtracting its infimum and normalizing gives a $[0, 1]$ -valued test function. The total-variation dual bound yields

$$\left| \int f_B d\sigma \right| \leq \eta(T) \|\sigma\|_{\text{TV}}.$$

Taking the supremum over B proves the result. $\square$

Consequences:

- positive descendant heredity requires assignment-distinguishable state at every complete screening bottleneck;
- it also requires nonzero downstream transmissivity;
- neither condition alone is sufficient;
- a complete matched transmission cut that erases the parent distinction gives zero descendant contrast;
- over multiple screened generations,

$$\delta_k \leq \delta_0 \prod_{j=0}^{k-1} \eta(T_j).$$

5.3 Theorem 8 — Aligned distributional heredity certificate

Let:

- $H0^*$ require $P(L = l_i \mid do(P = p_i)) = 1$, for distinct p_0, p_1 and distinct l_0, l_1 ;
- $H1^*$ establish a lineage-state intervention effect on the child;
- $H23^*$ match one complete joint context law μ across all four parent/lineage arms;

- $H5^\dagger$ require $C = F_H(L, W, E_H)$, no endogenous child parent outside (L, W) , one unchanged function and noise law λ_H , child noise independent of the complete upstream exogenous vector, and explicitly $E_H \perp W$ under every one of the four arms.

The arms are $G_i = do(P = p_i)$ and $I_i = do(L = l_i)$. All are surgical. Set

$$Q(c \mid l, w) = P_{\lambda_H}(F_H(l, w, E_H) = c), \quad Q_{l,\mu}(c) = \sum_w Q(c \mid l, w)\mu(w).$$

The full finite-state dictionary, definition of basic CLAIM-H, and proof are given in Mathematical Supplement S4.0. Matching the marginals of W and E_H separately cannot replace their required separation.

Then

$$\boxed{H0^* + H1^* + H23^* + H5^\dagger \implies \text{basic CLAIM-H.}}$$

The certified lineage-response separation is

$$\Delta_H = d_{TV}(\mathcal{Q}_{l_0,\mu}, \mathcal{Q}_{l_1,\mu}) > 0.$$

This is aligned **distributional causal transmission**. It does not assert identity copying, unit-level exclusive mediation, biological reproduction, or reflexive maintenance.

Within this certificate architecture, $H0^*, H1^*, H23^*$ are deletion-indispensable and $H5^\dagger$ is a load-bearing model-class boundary. Section 12 shows why such deletion results must not be promoted to global necessity.

6. Causal reflexive maintenance

6.1 Matched support effect

Let U_t be a physical output/support carrier whose support-relevant causal signature is assignment-mediated. Let S_U be a sham/control regime and B_U a matched block of the declared support coupling. Let

$$R_{t,\Delta} = \mathbf{1}\{\text{future state has operational-assignment capacity}\}.$$

The fixed future-assay capacity set is measurable under Section 2.0, so this indicator is a random variable under each comparison regime. S19 supplies sufficient diagnostic regularity; measurability is not inferred from an unrestricted existential intervention quantifier.

Define

$$\rho = P(R_{t,\Delta} = 1 \mid S_U) - P(R_{t,\Delta} = 1 \mid B_U).$$

Reflexive maintenance requires both assignment mediation of the support-relevant carrier and $\rho > 0$. Common resource covariance, generic housekeeping, and graph reachability alone are insufficient. For the fixed-state certificate in Section 6.5, this connection is imposed by RASM+: two matched, natural organization-setting regimes load the same distinct support states used in the sham/block capacity comparison. The positive capacity contrast then shows that this organization-dependent support difference is used by the maintenance mechanism. A change in an unrelated coordinate of the whole output is insufficient.

6.2 Theorem 9 — Reflexive-support bottleneck

Let H be a causal screening cut for every declared support path. Write

$$\mu_1 = \mathcal{L}(H \mid S_U), \quad \mu_0 = \mathcal{L}(H \mid B_U),$$

and require a common measurable downstream kernel T from the screening state to the future assay state. For the fixed measurable capacity event E_{cap} , set

$$r(h) = T(h, E_{\text{cap}}) \in [0, 1].$$

Then

$$\boxed{\rho = \int r(h)[\mu_1 - \mu_0](dh)}$$

and

$$\boxed{|\rho| \leq \eta_R \delta_H,}$$

where

$$\delta_H = \|\mu_1 - \mu_0\|_{\text{TV}}, \quad \eta_R = \sup_{h, h'} |r(h) - r(h')|.$$

The proof is the same total-variation oscillation argument as in the heredity bottleneck.

Thus positive reflexive support requires both an upstream support-state distinction and a downstream capacity sensitivity, with the correct signed alignment. If the matched comparison yields identical screening-state laws, $\mu_1 = \mu_0$, the common downstream kernel gives $\rho = 0$. The existence of a common downstream kernel alone does not imply zero effect.

6.3 Proposition 1 — Heredity and reflexive maintenance are non-equivalent

There are hereditary systems in which assignment-mediated outputs have no effect on future capacity, so heredity does not imply reflexivity.

Conversely, an assignment-mediated output can causally stabilize or replenish future assignment capacity while descendant orientation is independently redrawn, so reflexivity does not imply lineage heredity.

Therefore these are independent certification obligations conditional on operational assignment. Explicit finite witnesses, with their declared lineage and future-assay interfaces, are given in Mathematical Supplement S18.1.

6.4 Support-loop/re-entry certificate

The support-loop/re-entry certificate uses organization A , support state S , later organization A^+ , and matched context W . It defines a forward-active **support-response** class

$$\mathcal{A}_\cup(a_0, \mu) = \{a : \Sigma_{a, \mu} \neq \Sigma_{a_0, \mu}\}.$$

Under its printed conditions

$$R0^* + R1^* + R23^* + R5S^\dagger + R5A^\dagger + R6^\dagger,$$

it proves an aligned one-step support-loop/re-entry statement with

$$\Delta_R = M_{s_1, \mu}(\mathcal{A}_\cup) - M_{s_0, \mu}(\mathcal{A}_\cup) > 0.$$

$R6^\dagger$ gives standardized next-support re-entry for all certification-relevant later states.

Scope relative to future assignment capacity

Membership in $\mathcal{A}_\cup$ means “non-baseline support-response behavior.” It does **not** logically imply that the later state supports the specified operational-assignment assay.

Therefore the support-loop/re-entry theorem remains valid at its support-loop/re-entry target, but it is not by itself the full actual future-assignment-capacity certificate.

The Mathematical Supplement gives a four-state counterexample in which $\Delta_R > 0$ but the true future operational-assignment-capacity effect is zero.

6.5 Actual-future-state capacity-guarded reflexive maintenance

Capacity inferred from A^+ can be misleading if another coordinate of the actual future state prevents operational assignment. The certificate therefore requires the specified organization-level signature to agree with the interventional signature of the actual future state.

Let

$$C^+ \in \mathcal{C}^+$$

be the actual causal state at the specified future operational-assignment assay cut, and let

$$\mathcal{W}_{\text{OA}}^+$$

be the specified separating/generating family of future assay intervention words. Define

$$\Xi_{\text{OA}}^+(c) = (P_c^w)_{w \in \mathcal{W}_{\text{OA}}^+}.$$

Fix a predeclared map

$$\hat{\Xi} : \mathcal{A} \rightarrow \mathfrak{S}_{\text{OA}}^+$$

and define

$$\mathcal{A}_{\text{cap}} = \{a : \widehat{\Xi}(a) \text{ satisfies the fixed OA criterion}\}.$$

Equip the signature space with its declared sigma-field. Require the signature-capacity subset $\mathfrak{S}_{\text{cap}}^+$ to be measurable and the maps Ξ_{OA}^+ and $\widehat{\Xi}$ to be measurable. Thus $\mathcal{C}_{\text{OA}}^+ = (\Xi_{\text{OA}}^+)^{-1}(\mathfrak{S}_{\text{cap}}^+)$ and $\mathcal{A}_{\text{cap}} = \widehat{\Xi}^{-1}(\mathfrak{S}_{\text{cap}}^+)$ are measurable. Require also the signature-equality event in the following binding to be measurable. These conditions are part of this capacity certificate's domain; Part II, Section 1 and S19 give sufficient conditions and the probability transfer.

Actual-future-state binding

Under both matched support comparison regimes I_0^+, I_1^+ , require

$$\Xi_{\text{OA}}^+(C^+) = \widehat{\Xi}(A^+) \quad \text{a.s.}$$

and hence

$$\mathbf{1}\{C^+ \in \mathcal{C}_{\text{OA}}^+\} = \mathbf{1}\{A^+ \in \mathcal{A}_{\text{cap}}\} \quad \text{a.s.}$$

This excludes an omitted future capacity coordinate.

Assignment-related support condition

In addition to current operational assignment and organization dependence of the whole output, require **RASM+**. There must be distinct current assignment-bearing organizations a_0, a_1 and matched natural regimes N_j whose sole organization intervention is $do(A = a_j)$, with the same fixed challenge and unchanged output/support mechanisms. The intervention changes only the declared assignment organization; non-target physical causes of support belong to the complete matched context. This surgical interpretation is a hard hypothesis, not a consequence of output differences. The regimes satisfy

$$P(S = s_j \mid N_j) = 1, \quad j = 0, 1, \quad s_0 \neq s_1,$$

where s_0, s_1 are exactly the states used below. The complete context law is μ in both natural regimes. The same maintenance function, parent interface, noise law and noise/context separation used in the support arms also hold in these regimes. Consequently

$$\mathcal{L}(A^+ \mid N_j) = M_{s_j, \mu}.$$

The positive capacity margin below establishes that the differing naturally loaded support states have differing capacity responses. RASM+ is an additional, explicit sufficient condition for this fixed-state certificate; it is not inferred from organization dependence of the entire output and is not claimed necessary for every possible reflexive system. The output and support may be constitutive features of the same response; this does not assert an independently intervenable $O \rightarrow S$ edge.

Natural sham and matched block

Let N_1 be the unblocked natural support regime for the current operational-assignment-capable witness a_1 .

The sham arm I_1^+ must preserve the complete natural protected output-to-support law:

$$\mathcal{L}_{I_1^+}(O, S, W) = \mathcal{L}_{N_1}(O, S, W).$$

It may not clamp an artificially favorable support state that the current assignment-mediated output would not naturally load.

The block arm I_0^+ differs only by blocking/clamping the declared **post-output support coupling/state** after O has been generated, while preserving the challenge, output law, matched context, upstream mechanisms, and non-target downstream mechanisms.

State realization and capacity inequality

The certificate here is the fixed-support-state form in the Mathematical Supplement, Part II, Sections 5–8. In addition to preserving the natural sham law, its two arms must satisfy

$$P(S = s_1 \mid I_1^+) = 1, \quad P(S = s_0 \mid I_0^+) = 1.$$

Consequently the natural regime must itself load s_1 almost surely. Preserving a random natural support law alone does not establish this condition; an artificially clamped favorable sham is inadmissible.

The maintenance mechanism is $A^+ = F_A(S, W, E_A)$, with no endogenous parents outside S, W , the same function and exogenous law λ_A in both arms, and $E_A \perp W$ under each arm. These are the R5A[†] conditions

of the supplement. Together with fixed support realization and matched complete context they give, for each measurable event B ,

$$P(A^+ \in B \mid I_j^+) = \sum_w P_{\lambda_A}[F_A(s_j, w, E_A) \in B] \mu(w) = M_{s_j, \mu}(B), \quad j = 0, 1.$$

The sum denotes an integral against μ when the context is not discrete. The actual-future-state binding then transfers this identity to the capacity event. No fixed-support kernel substitution is asserted for a random-support comparison that fails the state-realization condition.

The matched complete-context condition is formally named

$$R23^+ : \quad \mathcal{L}(W \mid I_0^+) = \mathcal{L}(W \mid I_1^+) = \mu.$$

With the intervention-stable maintenance kernel

$$M_{s, \mu}(a^+) = \sum_w P_{\lambda_A}[F_A(s, w, E_A) = a^+] \mu(w),$$

require

$$\Delta_R^{\text{cap}} = M_{s_1, \mu}(\mathcal{A}_{\text{cap}}) - M_{s_0, \mu}(\mathcal{A}_{\text{cap}}) > 0.$$

Then

$$\begin{aligned} \rho_{\text{actual}} &= P(C^+ \in \mathcal{C}_{\text{OA}}^+ \mid I_1^+) - P(C^+ \in \mathcal{C}_{\text{OA}}^+ \mid I_0^+) \\ &= \Delta_R^{\text{cap}} > 0. \end{aligned}$$

Thus the **actual** future operational-assignment-capacity effect holds. Together with current assignment and RASM+, this establishes the fixed-state one-step reflexive-maintenance criterion. The displayed probability equality remains valid under its fixed-support, common-mechanism and future-binding premises even when RASM+ is absent; in that case it establishes a support effect without the additional assignment-related interpretation.

The support-loop R6† condition may additionally certify one standardized next-support re-entry step, but R6† is not a substitute for actual future-capacity binding.

7. Exact composition: why local certificates do not factorize

A set of valid local certificates need not describe one joint system. The certificates C_A, C_H, C_R can each hold while their shared context coordinates admit no common joint law. The specified identifications then prevent a coherent $A \rightarrow H \rightarrow R$ realization. Mathematical Supplement S6.3 gives an example that preserves every local certificate.

7.1 Baseline binding does not determine boundary continuation

Mathematical Supplement S6.1 gives a finite example with a fair bit Q and baseline states $X_A = P_H = L_H = C_H = X_R = Q$. The continuation equations are $S_R = X_R \oplus W_R$ and $A_R^+ = S_R$. A standardized context $W_R = 0$ preserves the bound bit, whereas the actual context $W_R = 1$ flips it. Conditional on each bound value $Q = q$, the two continuation laws have total-variation distance one.

The conclusion established by this example is

$$\boxed{\text{BASELINE STATE BINDING} \not\Rightarrow \text{CONDITIONAL BOUNDARY CONTINUATION MATCHING.}}$$

Binding is not preserved by every intervention: under $do(X_A = i)$, $P_H = Q$ remains unchanged and equals X_A with probability only 1/2. The example is therefore not a witness satisfying the full regime-wise binding block in S17, and no complete local R(loop) certificate is asserted for it. It isolates one boundary obligation. The positive construction in Part III establishes the stronger unified target using its full intervention atlas.

7.2 Other obstruction families

The following obstructions have distinct premise strengths. S6.2–S6.4 give kernel, probability-gluing, and causal-chart examples; S18.3 gives order-level examples:

- literal state/context compatibility does not prevent a **boundary-kernel selection shift**;
- pairwise overlap-law consistency does not imply existence of a **global joint law**;
- pairwise acyclic causal charts can require a **global causal cycle**;
- an assigned pair of epochs can violate a separately required handoff order;

- two individually acyclic order constraints can have a cyclic union.

The last two assertions concern ordering data; they do not assert independently complete A/H/R certificates for those two examples.

These results do not imply that one particular named sufficient block is globally necessary. They show failure modes that any sufficient unification theorem must exclude, neutralize, or otherwise account for when its ambient class contains the corresponding countermodels.

7.3 Exact unified target and strengthened capacity target

The exact target $CLAIM-AHR^*$ asks for one finite acyclic causal system in which:

1. the local A/H/R structural witnesses embed faithfully;
2. their required regime laws are realized;
3. every required positive-support handoff has an actual downstream joint continuation law equal to the reference target continuation law.

The support-loop/re-entry component in this target has the DFI-R(loop) support-loop/re-entry semantics.

For **future assignment-capacity** meaning, define the strengthened target

$$CLAIM-AHR_{cap}^* := CLAIM-AHR^* + \text{DFI-R(cap) actual-future-capacity claim.}$$

The strengthened target retains the support-loop requirement and adds the actual-future-capacity criterion. The added DFI-R(cap) component carries its measurable capacity sets, maps and binding event from Section 6.5; composition does not supply measurability missing from a local component. The finite common construction uses discrete sigma-fields.

7.4 Exact assembly theorem

The theorem below gives sufficient conditions for realizing the local certificates and their prescribed boundary responses in one finite acyclic causal system. Its requirements all use the same local witnesses, state identifications, intervention families, response laws, and causal order.

On one fixed common datum, assume the three local certificates and the printed global compatibility obligations:

- structural embedding/fidelity;
- required and combined-regime state binding;
- boundary/continuation regime closure;
- actual boundary-kernel invariance;
- complete reference-response-signature context compatibility;
- one globally coherent completion-closed acyclic surgical atlas/SCM;
- temporal embedding compatible with the same causal order.

Then

$$C_A + C_H + C_R + \text{global compatibility} \implies CLAIM-AHR^*.$$

The central continuation step is not a target restatement. It factors target equality into:

$$K^{\text{act}}(\theta, w) = g(\phi(\theta), w)$$

plus the response-weighted context identity

$$\sum_w g(\phi(\theta), w) \nu_\theta(w) = \sum_w g(\phi(\theta), w) \mu_{\phi(\theta)}(w).$$

The actual law of total probability then yields the reference target continuation law.

These conditions occupy a nontrivial middle ground:

- literal full context-law equality is stronger than necessary;
- arbitrary partial response-signature matching is insufficient;
- context matching alone does not prevent intervention-induced kernel selection.

Mathematical Supplement S6.1 isolates a baseline-binding/continuation mismatch; S6.2–S6.4 give kernel-selection, probability-gluing, and causal-cycle obstructions. Their stated premises are retained individually. None substitutes baseline binding for the full regime-wise binding required by this sufficient composition theorem.

7.5 Explicit positive non-vacuity witness

The finite construction in the Mathematical Supplement, Part III begins with the empty-matching genesis stage and uses the same M_0/M_1 assignment classes across the post-genesis $A \rightarrow H \rightarrow R$ handoffs.

The response-domain typing is

$$V_{\text{phys}} = \{0, 1\}, \quad B_{\text{resp}} = V_{\text{phys}} \cup \{\emptyset\},$$

with

$$\Gamma = S_2(U) \times S_2(V_{\text{phys}})$$

fixing the common null response.

A crucial structural distinction is that the diagnostic output and specified support state are not related by an extra causal parent edge. Instead,

$$O_R = \text{read}(A_R, 0)$$

is the diagnostic assignment-mediated output, while the fixed $R5S^\dagger$ support mechanism is directly

$$F_S(a) = \mathbf{1}\{\text{read}(a, 0) = 1\}, \quad S_R = F_S(A_R).$$

Thus $S_R = \mathbf{1}\{O_R = 1\}$ pointwise, but the fixed structural parent set of S_R is the declared organization/context interface rather than O_R .

Re-entry uses the exact same structural function:

$$S_{\text{next}} = F_S(A_R^+).$$

The margins of this construction are

$$\Delta_A = 1, \quad \Delta_H = 1, \quad \Delta_R = \frac{1}{2}, \quad \Delta_R^{\text{cap}} = 1.$$

For UCCE, the construction specifies every required regime through one fixed datum

$$\mathfrak{D}_{\text{toy}}$$

containing:

- explicit quotient maps and identity boundary maps on the M_0/M_1 classes;
- finite A/H and H/R persistent source-regime families;
- finite boundary-preserving continuation families;
- every required combined regime $r \oplus e$;
- singleton fixed and actual context laws;
- explicit H and R joint continuation response coordinates g ;
- the corresponding actual continuation kernels K^{act} ;
- exact response-weighted context equalities;
- literal structural embeddings;
- one fully controlled completion-closed global atlas;
- one common strict causal/time order.

The R support mechanism embedded into that global SCM is exactly

$$S_R = F_S(A_R),$$

so the fixed $R5S^\dagger$ parent/interface condition is preserved structurally rather than only in reduced-form probabilities.

The strengthened capacity comparison uses the actual-future-state binding and natural sham/matched block conditions.

Symmetry scope

Only the genesis stage is claimed assignment-neutral and Γ -equivariant. No global post-genesis Γ -equivariance or equal named-orientation persistence is claimed.

8. Finite-system robustness and claim-specific perturbation

8.1 Theorem 10 — Positive reset forbids permanent named orientation

If, while a system remains in a named orientation sector, the conditional probability of exit at the next epoch is bounded below by $r > 0$, then

$$P(\tau_\theta > T) \leq (1 - r)^T, \quad P(\tau_\theta = \infty) = 0, \quad E[\tau_\theta] \leq \frac{1}{r}.$$

Thus finite-horizon or metastable retention does not imply permanent named-orientation fidelity.

8.2 No universal finite- N lifetime scaling

If the one-step exit hazard lies between $\underline{q}_N$ and $\bar{q}_N$, then

$$(1 - \bar{q}_N)^T \leq P(\tau > T) \leq (1 - \underline{q}_N)^T.$$

No universal system-size scaling follows without an independent scaling hypothesis for the hazard sequence.

8.3 Label-blind error contracts channel contrasts

For a common Markov error kernel E ,

$$\|\mu E - \nu E\|_{\text{TV}} \leq \eta(E) \|\mu - \nu\|_{\text{TV}}.$$

For the convex mixture

$$E_\varepsilon = (1 - \varepsilon)I + \varepsilon R, \quad 0 \leq \varepsilon \leq 1,$$

with identical rows in R ,

$$\|\mu E_\varepsilon - \nu E_\varepsilon\|_{\text{TV}} = (1 - \varepsilon) \|\mu - \nu\|_{\text{TV}}.$$

There is no model-independent positive error radius over the entire model class because baseline diagnostic margins can be arbitrarily small.

8.4 Theorem 11 — Strict-margin openness

Suppose a finite theorem package is certified by diagnostics f_i with strict margins $m_i = f_i(P) > 0$. Assume each f_i is locally Lipschitz with constant L_i inside the admissible structural class and has a locality radius $r_i > 0$. Then every perturbation satisfying

$$d(P', P) < \varepsilon_* := \min_i \left\{ r_i, \frac{m_i}{L_i} \right\},$$

with $m_i/L_i = \infty$ when $L_i = 0$, preserves all strict inequalities.

This is an open-neighborhood theorem **inside the declared structural class**. Adding a hidden target-map-dependent transition is not a “small perturbation”; it leaves the admissible class.

8.5 Theorem 12 — Local parasite noninvasion

Let $d \geq 1$ be finite, and let F be defined on a relative neighborhood of zero in $\mathbb{R}_+^d$, with values in $\mathbb{R}_+^d$. The evolution is the local discrete iteration $y_{k+1} = F(y_k)$. Assume

$$F(0) = 0, \quad F(y) = Ly + o(\|y\|) \quad (y \rightarrow 0 \text{ in } \mathbb{R}_+^d),$$

where L is a finite $d \times d$ matrix with nonnegative entries. Then:

- $\rho(L) < 1$: some neighborhood of zero is forward invariant and all iterations started there converge to zero at a geometric rate in an equivalent norm;
- $\rho(L) > 1$: there is $v \in \mathbb{R}_+^d \setminus \{0\}$ with $Lv \geq \rho(L)v$, and $\|F(tv)\|_1 > \|tv\|_1$ for all sufficiently small $t > 0$;
- $\rho(L) = 1$: linearization alone does not determine decay, neutrality or growth.

Mathematical Supplement S20 proves all three branches, including forward invariance in the decay branch and critical scalar examples. The growth statement is a local first-order direction, not a claim of growth in every direction or of global invasion. No continuous-time or infinite-dimensional extension is asserted.

Parasite resistance is therefore an additional local stability property, not an automatic consequence of assignment, heredity, or reflexive support.

9. Unified finite-error and finite-sample certification

The exact UCCE theorem distinguishes causal meaning from probabilistic implementation. The finite-error layer preserves that distinction.

9.1 Hard structural and causal conditions

The following remain exact model-class/admissibility predicates:

1. fixed structural/model-class fidelity;
2. no-hidden-interpreter / no-hidden-fitness / no-bypass semantics;
3. surgical intervention target and meaning;
4. required-regime identity;
5. one common causal atlas/datum;
6. strict causal acyclicity and within-epoch causal order.

A wrong intervention or wrong causal structure is not converted into the intended one merely because its observed law is close in total variation.

9.2 Primitive soft-error quantities

Hard causal semantics remain exact. Quantitative error is tracked only after those gates pass. The cell, binding event, joint continuation response and transcript below are measurable. Their stated conditional laws are measurable kernels, and the TV defects integrated over context are measurable functions. These conditions, including measurable capacity-event indicators when tested, are required in a general-state extension; S19.3 explains their use.

For a required handoff cell z , define model-layer errors:

- s_z^{mod} : state-handoff mismatch;
- k_z^{mod} : continuation-kernel TV defect on matched state/context;
- c_z^{mod} : response-weighted context defect;
- ι_z : protocol/transcript-law TV defect;
- $h_z > 0$: a pre-specified positive support floor for the required conditioning cell.

The errors share one conditioning convention. For the model law conditional on the required cell z , let G be correct state binding and $\nu_z^G = \mathcal{L}(W \mid z, G)$. Then $s_z^{\text{mod}} = P(G^c \mid z)$. The kernel defect compares the actual conditional joint continuation kernel on (z, G, W) with the intended one, averaged under ν_z^G ; the context defect compares the intended joint continuation law under ν_z^G with its fixed reference-context law. Both defects refer to the matched branch, not to $\mathcal{L}(W \mid z)$. If $P(G \mid z) = 0$, use the bound one without defining an empty-branch kernel.

The transcript used for ι_z must determine the conditioning cell, state-binding event and full tested joint continuation response, or admit a common downstream kernel to that joint object. Its model reference cell has probability at least h_z , and its realized cell must have positive probability. Mathematical Supplement S8-S10 gives the precise definitions and proofs.

9.3 Structural-softening no-go

There are models whose required-regime probability laws are exactly equal to the reference laws while an off-regime intervention reveals an incorrect bypass/structural mechanism.

Therefore

$$\boxed{\text{required-law TV defect } 0 \not\Rightarrow \text{structural fidelity.}}$$

The Mathematical Supplement prints a binary $C = L$ versus $C = L \oplus Z$ witness.

9.4 Local margin stability

If implemented local comparison-arm laws differ from the reference A/H/R laws by TV errors $\eta_{X,0}, \eta_{X,1}$, then

$$\widetilde{\Delta}_X \geq \Delta_X - \eta_{X,0} - \eta_{X,1}, \quad X \in \{A, H, R\}.$$

For the strengthened capacity event,

$$\boxed{\widetilde{\Delta}_R^{\text{cap}} \geq \Delta_R^{\text{cap}} - \eta_{R,0} - \eta_{R,1}.$$

9.5 Primitive-to-realized continuation error transport

On one required handoff cell,

$$\delta_z \leq s_z^{\text{mod}} + (1 - s_z^{\text{mod}}) \min\{1, k_z^{\text{mod}} + c_z^{\text{mod}}\}.$$

The mismatch branch costs at most its probability s_z^{mod} . Conditional on a matched state, triangle inequality adds the kernel and response-weighted context defects.

Protocol/transcript mismatch is amplified when conditioning on a rare cell. Using the fixed support floor h_z , define

$$\sigma_z = \min\left\{1, s_z^{\text{mod}} + \frac{\ell_z}{h_z}\right\},$$

$$\beta_z = \min\left\{1, \frac{\ell_z}{h_z} + s_z^{\text{mod}} + (1 - s_z^{\text{mod}}) \min\{1, k_z^{\text{mod}} + c_z^{\text{mod}}\}\right\}.$$

Then the realized defects obey

$$s_z^{\text{real}} \leq \sigma_z, \quad \delta_z^{\text{real}} \leq \beta_z.$$

Thus the final continuation budget is **derived from lower-level errors** rather than simply assumed.

9.6 Unified finite-error envelope

The capacity-strengthened target in Section 7.3 retains the support-loop target and adds the capacity requirement. Accordingly, retain all four positive floors

$$0 < \underline{\Delta}_X < \Delta_X, \quad X \in \{A, H, R(\text{loop}), R(\text{cap})\}.$$

For each diagnostic, let P_{X0}, P_{X1} be the ideal arm laws and $\tilde{P}_{X0}, \tilde{P}_{X1}$ the implemented laws, with $d_{\text{TV}}(P_{Xi}, \tilde{P}_{Xi}) \leq \eta_{Xi}$. Impose the explicit erosion budget

$$e_X := \eta_{X0} + \eta_{X1} \leq \Delta_X - \underline{\Delta}_X.$$

The triangle inequality gives $d_{\text{TV}}(\tilde{P}_{X0}, \tilde{P}_{X1}) \geq \Delta_X - e_X$ for a TV diagnostic. For a fixed-event signed contrast, the two event-probability errors give the same lower bound. These arguments apply to each printed witness separately; they require no cancellation of errors between the loop and capacity events.

Assume the exact structural conditions, these local erosion bounds, and every required cell's transported bounds $\sigma_z \leq \bar{s}_z$, $\beta_z \leq \bar{\beta}_z$. Then the approximate capacity-strengthened unified target retains both the support-loop and actual-capacity margins together with the printed continuation budgets. When one common arm-error budget e_R is used for both R diagnostics, it must satisfy

$$e_R \leq \min\{\Delta_R - \underline{\Delta}_R, \Delta_R^{\text{cap}} - \underline{\Delta}_R^{\text{cap}}\}.$$

A separate corollary can replace the loop diagnostic by the capacity event and require only the capacity floor. That replacement target is not the strengthened conjunction $CLAIM-AHR_{\text{cap}}^*$ defined in Section 7.3. Implementation erosion is also distinct from the statistical confidence radii in Section 9.7.

9.7 Whole-envelope finite-sample certification

For a fixed $n \geq 1$, a finite alphabet of size m , independent identically distributed observations with common law P , and $\alpha \in (0, 1)$, the empirical law $\hat{P}_n$ has conservative total-variation radius

$$r_{\text{TV}}(n, m, \alpha) = \min\left\{1, \frac{m}{2} \sqrt{\frac{\log(2m/\alpha)}{2n}}\right\}, \quad P\{d_{\text{TV}}(\hat{P}_n, P) \leq r_{\text{TV}}(n, m, \alpha)\} \geq 1 - \alpha.$$

For a predeclared finite family of K arms/cells, choose positive budgets with $\sum_k \alpha_k \leq \alpha_{\text{infer}}$. Let θ denote the true vector of quantities and $\mathcal{C}(D)$ their joint confidence set derived from valid cellwise events. Its coverage event is

$$E_\theta = \{\theta \in \mathcal{C}(D)\}, \quad P_\theta(E_\theta^c) \leq \alpha_{\text{infer}}.$$

Let $\mathcal{T}$ be the declared measurable quantitative target envelope within the admissible structural model class. Coverage, adequate acquisition and the implemented PASS decision are measurable events of the data. For general data spaces, measurability of a set-inclusion decision is an explicit requirement; it is not inferred solely from writing $\mathcal{C}(D) \subseteq \mathcal{T}$. The finite-alphabet designs in S13 meet these requirements as shown in S19.4. Define PASS only when the required hard conditions have been established and $\mathcal{C}(D) \subseteq \mathcal{T}$.

The soundness implication is

$$P_\theta(\text{PASS} \wedge \theta \notin \mathcal{T}) \leq \alpha_{\text{infer}}.$$

This bound concerns the quantitative certificate under valid hard-condition assessment. Statistical precision cannot certify an unverified structural condition or absorb its failure.

Coverage and acquisition are distinct from certification power

Let A be the predeclared adequate-acquisition event, with $P_\theta(A^c) \leq \alpha_{\text{acq}}$. No independence between acquisition and coverage is needed for

$$P_\theta(A \cap E_\theta) \geq 1 - \alpha_{\text{acq}} - \alpha_{\text{infer}}.$$

This is an acquisition-and-coverage probability, not a lower bound on PASS. Even when $\theta \in \mathcal{T}$, a confidence set may cross the target boundary and yield INCONCLUSIVE. A power claim additionally requires a quantitative separation and precision condition ensuring that every record in $A \cap E_\theta$ satisfies all PASS criteria.

One explicit sufficient condition is given in the Mathematical Supplement, S13.2. Suppose the soft requirements are $g_\ell(\theta) > 0$, their true margins are at least $\gamma_\ell > 0$, and the functions have declared coordinatewise TV Lipschitz constants $L_{\ell k}$ on the relevant parameter domain. If acquisition ensures confidence-ball radii no larger than r_k , and

$$2 \sum_k L_{\ell k} r_k < \gamma_\ell \quad \text{for every } \ell,$$

then coverage puts the entire confidence set inside the target. When the hard conditions are also established on that event, this explicit subclass satisfies $P_\theta(\text{PASS}) \geq 1 - \alpha_{\text{acq}} - \alpha_{\text{infer}}$. No uniform power guarantee is asserted for all systems with merely positive margins.

Realized counts and stopping

Using a realized matched count in a fixed-sample radius requires a valid sampling argument. For example, in a fixed number of independent identically distributed trials with predeclared cell tags, conditioning on the complete tag vector leaves the within-cell observations independent with their common conditional law; averaging these conditional bounds validates the radius at the resulting random count. Empty required cells are INCONCLUSIVE. Selection based on the responses or adaptive stopping does not inherit this argument. S13.3 specifies the sampling conditions and acquisition event.

An outcome-dependent stopping rule requires simultaneous-in-time coverage or a separately proved calibration for that rule. Preregistration alone does not supply either. An elementary option for an IID cell stream is to use, at each $n \geq 1$,

$$\alpha_{k,n} = \frac{\alpha_k}{n(n+1)}, \quad r_{k,n} = r_{\text{TV}}(n, m_k, \alpha_{k,n}).$$

The sum over cells and times is at most α_{infer} , so a union bound gives simultaneous coverage and permits stopping within these verified streams. The original fixed- n radius with an unchanged budget is not substituted at a data-dependent time. The Mathematical Supplement, S13.4, gives the proof; the Experimental Methods Supplement, E3.2 and E3.5, binds the protocol to one valid sampling design.

10. Abstract resource admissibility

Resource feasibility is an added accounting layer. It neither creates assignment nor substitutes for causal maintenance.

10.1 No canonical universal resource scalar

The causal-interventional path law alone does not single out one scalar called resource, free energy, or entropy production. Two conservative extensions can preserve the same projected causal path laws while appending different non-feedback bookkeeping coordinates.

Therefore resource claims require an explicitly declared accounting structure.

10.2 Theorem 13 — Cancellative prefix theorem

Let $(M, +, 0)$ be a commutative cancellative resource monoid. Let B_t be sequential residual stock, I_t inflow, and C_t requirement, with causal balance

$$B_t + I_t = C_t + B_{t+1}.$$

Fix the prefix convention

$$C^{[t]} = \sum_{u=0}^{t-1} C_u, \quad I^{[t]} = \sum_{u=0}^{t-1} I_u, \quad C^{[0]} = I^{[0]} = 0.$$

Then finite-horizon sequential resource feasibility is equivalent to aggregate prefix decomposability

$$C^{[t]} \preceq_M B_0 + I^{[t]} \quad \forall t.$$

For sufficiency, choose residuals R_t with

$$C^{[t]} + R_t = B_0 + I^{[t]}.$$

Consecutive prefixes give

$$C^{[t]} + R_t + I_t = C^{[t]} + C_t + R_{t+1},$$

and cancellation yields

$$R_t + I_t = C_t + R_{t+1}.$$

Thus $B_t = R_t$ is a sequential path.

The resource analysis explicitly does **not** claim cancellation is universally necessary.

10.3 Conditional resource subclasses

Stronger conclusions require stronger structures:

- ordered-group/cone structure for scalarization or geometric bounds;
- explicit capacity and chronology for finite storage;
- process hypotheses for stochastic survival;
- a positive-cost/dissipative axiom for sustained-flux necessity.

In particular, persistent causal maintenance does not imply strictly positive resource dissipation without an additional positive-cost axiom. A zero-cost abstract maintained state is a valid countermodel to that unconditional inference.

11. Exact resource minimality: the Causal Residual Kernel

This section gives an exact characterization in terms of residual viability.

11.1 Residual viability

Let $(M, +, 0)$ be a commutative monoid and define

$$x \preceq_M y \iff \exists r \in M : x + r = y.$$

A residual state is

$$(a, b, r) \in \Omega \quad \text{with} \quad a + r = b.$$

Here a is cumulative demand, b cumulative available supply, and r the current sequential residual.

A current move (c, i) is aggregate-admissible when

$$a + c \preceq_M b + i.$$

A causal successor residual s must satisfy

$$r + i = c + s.$$

A set $W \subseteq \Omega$ is **causally residual-invariant** when every state in W , against every currently aggregate-admissible move, has at least one successor whose next residual state remains in W . Let $W_{\max}$ be the union of all such invariant sets.

11.2 Theorem 14 — Exact CRK characterization

The following are equivalent:

1. for every initial stock B_0 , there exists a nonanticipating residual policy that maintains sequential

- balance against every stream whose aggregate prefix constraints remain feasible;
2. $(0, b, b) \in W_{\max}$ for every $b \in M$;
 3. for every $a \preceq_M b$ there is a nonempty viable residual subfiber
$$\mathcal{V}(a, b) \subseteq \{r : a + r = b\}$$

such that

$$r \in \mathcal{V}(a, b), \quad a + c \preceq_M b + i$$

implies existence of

$$s \in \mathcal{V}(a + c, b + i)$$

satisfying

$$r + i = c + s.$$

This equivalent property is **CRK completeness**.

Selection semantics

The abstract theorem is set-theoretic: from residual invariance, one chooses one legal successor for each current state/move pair. For finite monoids this is an ordinary finite selector.

For general measurable state spaces, any claim that the selected policy is **measurable** requires an additional measurable-selection hypothesis. CRK completeness by itself is not presented as a measurable-selection theorem.

Proof

(1) $\Rightarrow$ (2). Fix b . Collect all residual states reachable under a winning nonanticipating policy from initial state $(0, b, b)$ over all finite adversarial histories. At any reachable state and any current aggregate-admissible move, winning against the continuation requires at least one legal successor that is again reachable. Hence the reachable set is residual-invariant and lies in $W_{\max}$, so $(0, b, b) \in W_{\max}$.

(2) $\Rightarrow$ (1). At each state in $W_{\max}$, residual invariance supplies a successor for every current admissible move. Choosing one such successor yields a memoryless nonanticipating winning policy. No future stream is consulted.

(2) $\Rightarrow$ (3). For any $a \preceq_M b$, apply current move $(a, 0)$ from $(0, b, b)$. Invariance yields at least one r with $a + r = b$ and $(a, b, r) \in W_{\max}$. Define

$$\mathcal{V}(a, b) = \{r : (a, b, r) \in W_{\max}\}.$$

Residual invariance gives the cross-fiber closure.

(3) $\Rightarrow$ (2). Let

$$W_{\mathcal{V}} = \{(a, b, r) : r \in \mathcal{V}(a, b)\}.$$

The cross-fiber condition makes $W_{\mathcal{V}}$ residual-invariant. For $a = 0$, the identity law forces the unique residual $r = b$, so $(0, b, b) \in W_{\mathcal{V}} \subseteq W_{\max}$. $\square$

11.3 Algebraic sufficient branches

- Every cancellative commutative monoid is CRK-complete.
- Cancellation is not necessary: $(\{0, 1\}, \max)$ is noncancellative and CRK-complete.
- Every commutative idempotent monoid is CRK-complete: if $a \preceq b$, choose residual $r = b$; after an admissible move choose $s = b + i$.
- A commutative monoid that is both cancellative and idempotent is trivial, so the two sufficient branches are genuinely distinct.
- A Coherent Residual Selector (CRS), defined below, is sufficient for CRK completeness. No converse is asserted in this article.
- CRK is invariant under monoid isomorphism but is not preserved under arbitrary surjective homomorphic quotient.

A **Coherent Residual Selector** on a commutative monoid M is a function $\delta(a, b)$ on every pair $a \preceq_M b$ such that

$$a + \delta(a, b) = b,$$

and, for every $c, i \in M$ with $a + c \preceq_M b + i$,

$$\delta(a, b) + i = c + \delta(a + c, b + i).$$

To prove sufficiency, let $W_{\delta} = \{(a, b, \delta(a, b)) : a \preceq_M b\}$. The first identity puts it inside Ω ; the second

provides a successor in the same set for every admissible current move. It is therefore residual-invariant. Since $0 + \delta(0, b) = b$, it contains every root $(0, b, b)$. Theorem 14 gives CRK completeness. This is a set-theoretic statement; it adds no measurable, continuous or computable selection guarantee.

For clarity, failure under arbitrary surjective quotient has a concrete witness. Map the additive monoid $\mathbb{N}$ onto $\{0, 1, 2\}$ by $f(n) = \min(n, 2)$, with truncated addition in the target. From initial stock 1, a first demand/inflow pair $(1, 0)$ forces residual 0. A next pair $(2, 1)$ is cumulatively feasible in the target, since both cumulative totals truncate to 2, but available residual plus inflow is 1 and cannot equal demand 2 plus any residual. The source is cancellative and prefix-complete; the target is not.

Within the declared commutative-monoid residual game, CRK is the exact universal no-foresight prefix-completeness condition.

12. Minimality, necessity, and theorem universality

12.1 Proposition 2 — Why unrestricted “weakest antecedent” is degenerate

Fix an ambient model-class predicate A and target claim C . If C itself is allowed as an antecedent, then C is trivially a least sufficient formula. If two mechanisms H_1, H_2 each imply C , then the disjunction $H_1 \vee H_2$ is weaker than either branch while still sufficient.

Therefore logic alone cannot supply a scientifically privileged premise language. Foundational minimality must be evaluated relative to a declared noncircular lower-level set of admissible assumptions.

12.2 Theorem 15 — Semantic basis poset

Let

$$\mathcal{H} = \{h_1, \dots, h_m\}$$

be a fixed finite set of admissible assumptions. For $S \subseteq \mathcal{H}$, let

$$H_S = \bigwedge_{h \in S} h.$$

Define semantic equivalence inside the ambient class by

$$S \equiv_A T \iff A \models (H_S \leftrightarrow H_T).$$

Let $[S]_A$ denote the equivalence class and order the classes weak-to-strong by

$$[S]_A \preceq_A [T]_A \iff A \wedge H_T \models H_S.$$

The sufficient-basis family is

$$\text{Suff}_{A,C}(\mathcal{H}) = \{S \subseteq \mathcal{H} : A \wedge H_S \models C\}.$$

After quotienting by $\equiv_A$, sufficient classes form a partially ordered set and an upper set in the weak-to-strong order.

For a basis S , define its theorem domain

$$\mathcal{U}_A(S) = \text{Mod}(A \wedge H_S).$$

Then

$$[S]_A \preceq_A [T]_A \iff \mathcal{U}_A(T) \subseteq \mathcal{U}_A(S).$$

Hence

minimal sufficient class $\iff$ inclusion-maximal sufficient conjunction-generated theorem domain.
--

No maximality is claimed over arbitrary Boolean formulas unless that larger premise language is explicitly admitted and quotiented.

12.3 Library expansion

Adding only definitional aliases equivalent to old-library conjunctions does not change the semantic poset. Adding a genuinely weaker sufficient condition can change the set of minimal bases. Minimal sufficient bases are therefore relative to the chosen set of admissible assumptions, unless a separate necessity theorem for the ambient model class rules out all weaker alternatives.

12.4 Types of necessity and sufficiency

An assumption can be definitional, necessary throughout an ambient model class, part of the model-class boundary, indispensable within a particular sufficient construction, or one replaceable sufficient condition among others. An exact characterization establishes necessity and sufficiency for its stated target. A change of representation can leave the underlying property unchanged. These are different logical roles, not additional object-level theorems.

A counterexample obtained by deleting one assumption from a particular construction shows that the construction needs that assumption. It does not establish necessity throughout the ambient model class, namely

$$A \wedge C \models h.$$

A universal necessity claim therefore requires a separate proof with the corresponding model-class quantifiers.

12.5 Theorem 16 — Representation and conjunction-generated universality

If a bijection between ambient model classes preserves and reflects:

1. ambient membership;
2. the target claim;
3. the truth of every declared library atom under a bijective atom transport,

then it induces an order isomorphism between the corresponding semantic sufficient-basis posets.

Thus minimal bases and their maximal conjunction-generated theorem domains are representation-invariant under the declared semantic transport.

13. Non-implications and converse failures

Table 1. Non-implications and converse failures.

Tempting inference	Mathematical status
high correlation $\Rightarrow$ operational assignment	false; intervention-sensitive organization dependence is required [Mathematical Supplement, S18.2(a)]
neutral ensemble $\Rightarrow$ every realization is symmetry-fixed	false [Mathematical Supplement, S18.2(b)]
exactly fixed realized origin + deterministic	false [Theorem 1]
equivariance $\Rightarrow$ genesis	
state symmetry breaking $\Rightarrow$ assignment	false [Mathematical Supplement, S18.2(b)]
quotient-erased state $\Rightarrow$ contingent assignment	false [Theorem 2]
passive I/O quality $\Rightarrow$ decoder-free provenance	false [Theorem 5]
arbitrary adaptive input control alone $\Rightarrow$ decoder-free provenance	false on the Theorem 6 witness class
parent-child agreement $\Rightarrow$ causal heredity	false [Mathematical Supplement, S18.2(c)]
transmitted nuisance information $\Rightarrow$ assignment heredity	false [Mathematical Supplement, S18.2(d)]
graph path/cut alone $\Rightarrow$ causal screening	false [Mathematical Supplement, S18.2(e)]
heredity $\Rightarrow$ reflexive maintenance	false [Mathematical Supplement, S18.1(a)]
reflexive maintenance $\Rightarrow$ lineage heredity	false [Mathematical Supplement, S18.1(b)]
generic self-maintenance $\Rightarrow$ assignment-reflexive maintenance	false [Mathematical Supplement, S18.2(f)]
output/future-capacity covariance $\Rightarrow$ reflexivity	false [Mathematical Supplement, S18.2(g)]
baseline state binding $\Rightarrow$ conditional boundary continuation matching	false [Mathematical Supplement, S6.1]; no full regime-wise binding or local R certificate is asserted
pairwise probability compatibility $\Rightarrow$ global gluing	false [Mathematical Supplement, S6.3]
pairwise acyclic causal charts $\Rightarrow$ global acyclic SCM	false [Mathematical Supplement, S6.4]
law TV defect 0 $\Rightarrow$ structural correctness	false [Mathematical Supplement, S7]
finite-horizon retention $\Rightarrow$ permanent named orientation	false [Mathematical Supplement, S18.2(h)]
small error $\Rightarrow$ universal positive robustness radius	false [Mathematical Supplement, S18.2(i)]

Tempting inference	Mathematical status
assignment/reflexivity $\Rightarrow$ parasite immunity	false [Mathematical Supplement, S18.2(j)]
open resource supply $\Rightarrow$ causal maintenance	false [Mathematical Supplement, S18.2(k)]
persistent maintenance $\Rightarrow$ positive dissipation	false without a positive-cost axiom
aggregate prefix decomposability $\Rightarrow$ sequential resource path in every commutative monoid	false
cancellation $\Rightarrow$ resource prefix completeness	true, sufficient
cancellation is universally necessary	false
coarse resource feasibility $\Rightarrow$ fine resource feasibility	false without lifting assumptions [Mathematical Supplement, S18.2(l)]
deletion-indispensable premise $\Rightarrow$ global necessity	false [Mathematical Supplement, S18.2(m)]
one sufficient construction $\Rightarrow$ unique weakest axiom set	false / ill-posed without premise-language restriction

14. Logical dependencies

The order of exposition is not an implication chain. Assignment and origin definitions determine the relevant property; the identifiability no-go results specify what a restricted protocol cannot determine. Organization, heredity, and reflexive-maintenance certificates then impose separate structural and interventional conditions. Exact composition requires those certificates to share one compatible causal system, and finite-error certification applies only within the admissible structural class. Robustness, resource accounting, and premise-relative minimality constrain additional dimensions.

In particular, the support-loop target and actual-future-capacity target remain different. Adding the capacity certificate to an exact composition witness strengthens its interpreted conclusion; it does not change the meaning of the support-loop theorem. A chemical test must declare which target and which hypotheses it is assessing.

15. Experimental correspondence and certification

15.1 Mathematical truth, correspondence, and evidence

15.1.1 Mathematical specification and experimental correspondence

Let F denote the mathematical specification: the model class, assumptions, target properties, and conclusions to be assessed. Let $E(F)$ denote an experimental specification that identifies the proposed physical counterparts, interventions, observables, controls, and decision rules. These are specified independently of the confirmatory outcomes.

An experiment assesses whether the physical system and its interventions meet the assumptions of F , and whether the predicted observations occur. Agreement with a prediction cannot substitute for a mathematical proof. Failure of a proposed physical correspondence does not by itself refute the conditional theorem. A mathematical counterexample or a defect in the assumptions or proof is a separate issue.

15.1.2 Three logically different questions

We distinguish:

1. **Mathematical status:** what follows from the declared axioms/model class?
2. **Reality membership:** does the physical system in fact satisfy the required hypothesis class and intervention semantics?
3. **Empirical assessment/outcome:** does the available protocol and record adequately determine the relevant membership/discriminator, and what was observed?

Historical relevance is a fourth question and is kept separate.

This distinction blocks two common invalid inferences:

experiment matches consequence $\nRightarrow$ theorem proved,

and

experiment fails proposed correspondence $\nRightarrow$ mathematical theorem false.

The latter conclusion becomes available only if an independent mathematical counterexample, inconsistency, or proof defect is found.

15.2 The correspondence specification

15.2.1 Experimental specification

The Empirical Correspondence System (ECS) translates a fixed mathematical specification into an experimental assessment. Its empirical corroboration scan is the literature-comparison component, rather than the whole certification system. The logical sequence is:

mathematical specification $\rightarrow$ correspondence ontology $\rightarrow$ experimental discriminator $\rightarrow$ membership assessment $\rightarrow$ evidence classification.

The specification contains the following elements:

- **Mathematical target** records theorem object, hypotheses, claim class, discriminator, and testability.
- **Correspondence ontology** binds mathematical states, kernels, interventions, lineage variables, future-capacity variables, resource variables, and error quantities to physical candidates before outcomes are interpreted.
- **Experimental specification** states the hypotheses, interventions, observables, controls, exclusion criteria, tolerances, and predeclared decision rules.
- **Existing evidence** classifies already-published evidence without upgrading analogy into replication.
- **Hypothesis-class membership** checks origin, boundary, intervention, nuisance/context, causal order, resource class, and error class.
- **Composition** asks whether local assignment/heredity/reflexive certificates inhabit one compatible causal system.
- **Finite error and sampling** separates exact structural requirements from bounded implementation deviations and statistical uncertainty.
- **Robustness and resources** specifies the property, model class, admissible perturbations, and sequential resource balances. Thermodynamic cost requires an additional physical model.
- **Prospective replication** requires a preregistered protocol specifying the mathematical target, physical correspondence, implementation, controls, tolerances, and analysis before confirmatory data are collected.
- **Evidence classification** separates theorem status, real-system eligibility, existing correspondence, prospective replication, and historical relevance.

15.2.2 Model-class membership, assessment, and empirical evidence

Let S denote the physical system and F the specified mathematical and correspondence target. At the model level, reality membership is a property of the system:

$$\text{Member}(S, F) \in \{0, 1\},$$

where membership means that the complete physical system, boundary/resource class, and intervention semantics satisfy the stated model-class assumptions. This truth value is not created by the dataset.

The experimental record D supports a separate assessment state:

$$\text{Assess}(D, S, F) \in \{\text{supported}, \text{partial}, \text{violated}, \text{undetermined}\}.$$

An undetermined assessment is epistemic: it means the available record cannot determine the required discriminator or membership claim. It does not mean that the underlying system is physically non-member.

Separately define an evidence classification

$$\text{Ev}(D, S, F) \in \{\text{independent_confirmation}, \text{contextual_support}, \text{empirical_tension}, \text{nondiagnostic}, \text{incomparable}\}.$$

A system may strongly support a component mechanism yet fail to belong to the decoder-free-origin hypothesis class. An inadequate protocol can instead leave membership unresolved. Failure of membership and inability to assess membership are different outcomes. Neither component support nor an unresolved assessment establishes that the full system meets the specified assumptions.

For evidence to count as independent confirmation, it must not have been used to select or revise the correspondence map, discriminator, threshold, implementation, or analysis rule. The order of specification and data use must be documented. Evidence that influenced these choices provides contextual support,

not independent confirmation.

15.2.3 Certification, rejection, and inconclusive results

Each experimental test must specify its decision rules before the confirmatory outcomes are examined.

A structural violation means that the tested implementation does not satisfy a required model-class assumption. Certification (the PASS event used in the statistical results) requires the prespecified diagnostic criteria and all required structural and quantitative conditions. A decisive failure of a prespecified diagnostic or an established structural violation rejects the particular implementation claim. An inconclusive result means that the protocol cannot evaluate the required diagnostic, for example because matched sample counts, tracking, controls, or intervention support are inadequate.

A structural violation, insufficient evidence, and successful certification are different outcomes. An inconclusive result is not a weaker form of certification and cannot be converted into certification by deleting unsuccessful cells after inspecting their outcomes.

15.3 Mapping the theorem families to experiments

Table 2 links each mathematical result to an experimental question, the response or structural requirement to assess, and a characteristic violation of its requirements or scope. It is a correspondence index; the formal criteria remain in the definitions and theorems.

Table 2. Correspondence between mathematical results and experimental questions.

Mathematical result	Empirical question	Required signature	Characteristic violation
Operational assignment (§2)	Does the system satisfy the specified operational-assignment criterion rather than merely exhibit an unavoidable direct transformation?	the challenge-response relation satisfies the specified operational-assignment and contingency conditions	fixed direct chemistry relabeled as assignment
Origin neutrality (§2.4)	Was a decoder-equivalent relation absent in the declared origin ontology?	structural symmetry/equivariance, independently justified DC0 and the state-indexed nonconstructive diagnostic repertoire	a preinstalled relation; missing diagnostic completeness leaves assessment inconclusive
Genesis (§3)	Can symmetry-related mappings arise under one neutral law?	multiple admissible orientations without target-dependent enrichment	selection retains a preferred mapping
Theorems 5–6	Can the stated observation/intervention class identify provenance?	identifiability assessment for the specified protocol	passive/input-only evidence overclaims hidden provenance
DFI-A	Does later internal organization causally carry the relation?	matched organization-setting interventions separate output laws	intervention bypasses organization or changes nuisance
Heredity / DFI-H (§5)	Is assignment organization causally inherited rather than reconstructed?	parent intervention affects descendant assignment; identical screening-state laws give zero descendant contrast under the common downstream kernel	common-environment re-emergence
DFI-R(cap) (§6.5)	Do assignment-mediated outputs maintain actual future assignment capacity?	natural sham versus post-output block changes future operational-assignment capacity	block destroys unrelated capacity or omits future-state assay

Mathematical result	Empirical question	Required signature	Characteristic violation
UCCE	Do local A/H/R certificates compose in one system?	common state, context, kernel, intervention, gluing and time-order compatibility	local certificates cannot be globally glued
UFE	Does finite implementation preserve the certified causal semantics?	exact structural conditions plus preregistered soft-error budgets	metric closeness used to excuse wrong structure
Robustness (§8)	Is the specific property robust within the declared model class under the declared perturbation/error/reset conditions?	property- and model-class-indexed margin/lifetime/error assay	robustness in one property/model class extrapolated outside that class
Resources (§10)	Is the claimed maintenance resource-admissible under an explicit physical bookkeeping model?	explicit sequential material/resource accounting	hidden material/energy creation; causal/resource feasibility misread as identification of thermodynamic cost
CRK (§11)	Can a nonanticipating residual policy maintain sequential feasibility against every aggregate-admissible stream in the declared commutative-monoid class?	residual causal feasibility after each current move	aggregate budget substituted for sequential realization
Minimality (§12)	Which sufficient conjunctions are minimal within the declared ambient class and set of admissible assumptions?	assumption comparison within that fixed premise language	architecture-relative deletion results extrapolated to universal necessity

The robustness claim is therefore indexed jointly by the property and the declared model class; no cross-model-class extrapolation is licensed. Resource feasibility does not identify thermodynamic cost from causal structure alone: any energetic/dissipative cost claim requires an independently specified physical bookkeeping model. Minimality/necessity statements remain scoped to the declared ambient class and set of admissible assumptions and are not claims of a unique globally weakest named axiom set.

Two distinctions are central.

First, DFI-R(cap) concerns the effect of an assignment-mediated support path on actual future assignment capacity. It requires a matched intervention on the output-to-future-capacity route and evaluates the actual future state using a fixed future-assay interpretation. Abundance correlations alone do not answer this causal question.

Second, the UCCE certificate requires compatible state bindings, causal semantics, and joint continuation laws within one causal system. Individually plausible assays do not establish that compatibility. UFE bounds quantitative errors while retaining the structural causal requirements. Statistical closeness does not repair a structural mismatch.

15.4 Hard requirements and quantitative error

15.4.1 Structural conditions

A structural condition specifies what kind of physical system or intervention is being tested. Examples include:

- no pre-existing pair-specific decoder-equivalent relation at the origin;
- a challenge and response that remain distinct physical roles;
- an organization intervention that does not directly write the output;
- a parent intervention and transmission cut with the declared causal meaning;

- a future-capacity intervention that does not destroy unrelated capacity;
- one common causal atlas for a conjunction claim;
- preserved regime identity and causal/time order.

If a structural condition fails, the result is not repaired by increasing sample size.

15.4.2 Soft quantities

Only after the structural membership conditions are satisfied do quantitative implementation deviations become meaningful. Examples include:

- bounded pre-link leak;
- local response-margin loss;
- handoff mismatch;
- continuation approximation error;
- tracking error;
- positive-time clock error;
- finite observation error.

Let $e = (e_1, \dots, e_q)$ denote the predeclared soft-error vector and let $\mathcal{E}$ be the admissible error region. Certification requires

$$e \in \mathcal{E}$$

in addition to all structural conditions.

This separation prevents the invalid inference

$$\text{small distance between observed laws} \Rightarrow \text{correct causal implementation.}$$

16. Existing evidence and prospective validation

16.1 Selected evidence and its limits

The literature examples considered here were selected through August 2026 and do not constitute a systematic review. They illustrate component feasibility and the difference between an observed mechanism and membership in the full hypothesis class. In this selected set, no system is shown here to satisfy the complete conjunction. This is neither a nonexistence theorem nor a claim about all published work. The detailed discussion appears in the Experimental Methods Supplement, E1.

Table 3. Selected component precedents and their evidential limits.

Component precedent	What it informs	What remains unestablished here
Catalytic DNA templating [19]	Sequence-specific propagation and turnover	Decoder-free origin when recognition domains are designed in advance
Open-ended RNA replication [18]	Replication and environmental cycling	Contingent assignment and assignment-specific causal heredity
Autocatalytic RNA networks and ribozyme assembly [17,25]	Network support and persistence	An assignment-mediated causal effect on actual future assignment capacity
RNA-peptide and aminoacylation chemistry [21,26]	Support chemistry and cross-domain coupling	Neutral contingent genesis and the full causal certificate
Compartment cycling [27]	Repeated redistribution and replication	Assignment-specific heredity under the required intervention

A complete realization cannot be inferred by combining these examples across incompatible systems. The composition obligations of Section 7 apply to one common causal system, including its state bindings, continuation laws, intervention semantics, and time order.

16.2 Calibration must precede confirmation

The detailed protocol in the Experimental Methods Supplement, E3, distinguishes exploratory global engineering, preregistered non-contingency validation, and a fixed implementation eligible for a confirmatory mapping test. Global transducer, linker, challenge, readout, ablation, buffer, and timing parameters may

be developed in non-contingency systems and then held common across copies and pairings. Pair-specific mapping optimization is inadmissible.

Before validation, the protocol specifies the experimental unit, sample size, batch allocation, randomization, stopping, missing-data and exclusion rules, and any sequential design. Unplanned extension after inspecting outcomes invalidates the stated fixed-sample validation guarantee. Repeated frames or correlated droplets do not become independent observations merely by being counted separately. Fixed-sample bounds apply only under their stated sampling assumptions; all inferential allocations cover the complete declared family. The default design analyzes once at its declared fixed sample size. An outcome-dependent stopping alternative requires a proved simultaneous-in-time or stopping-specific error guarantee, not pre-registration alone. Adequate acquisition and confidence coverage do not imply PASS power without the separation and precision conditions of Section 9.7 and the Mathematical Supplement, S13.2–S13.4.

The validation matrix tests unlinked background, linked function, off-output routing, off-input challenge spillover, organization ablation, tracking perturbation, and aggregation. These controls are necessary but not sufficient to exclude artifacts that occur only in the full 2×2 context. A prospective confirmatory test also requires predeclared joint-context controls for higher-order interactions and bypass mechanisms. Failed or missing required controls cannot be repaired by post-hoc exclusion.

16.3 Conditional RNA-routing design

The Experimental Methods Supplement, E2–E5, examines exchangeable input and output copies joined by generic coupling chemistry with link-local transduction. Within-domain chemical identity is insufficient by itself: symmetry is a property of the complete preparation and joining protocol. Individual tracking and challenge must use an admissible non-mapping addressability layer that neither chooses the pairing nor writes the output.

The worked split-hammerhead routing design draws on distinct component precedents [20,22–24,35], whose demonstrated functions and architectural limits are distinguished in the Experimental Methods Supplement, E4. It remains a conditional design. Exact molecular performance, leakage, crosstalk, tracking perturbation, ablation specificity, multivalency, and batch variability have not been measured for the proposed construct. Its implementation is therefore not yet fixed for a confirmatory random 2×2 mapping experiment. No laboratory result or theory-conformant replication is reported.

Even a successful initial assignment test would not establish the entire conjunction. Assignment-specific heredity, future-capacity maintenance, single-system composition, finite-error control, robustness, and explicit resource accounting remain separate experimental obligations.

17. Scope, limitations, and related work

17.1 Scope and limitations

The mathematical results presented here do **not** establish:

- a specific chemical implementation;
- prebiotic plausibility;
- early-Earth history;
- origin of the Standard Genetic Code;
- identification of abstract roles with RNA, peptide, amino acid, tRNA, aaRS, ribosome, codon, or anticodon;
- a second complete reflexive-maintenance cycle;
- recursive multi-cycle reflexive closure;
- indefinite persistence;
- a universal positive error threshold;
- universal parasite immunity;
- universal finite-size lifetime scaling;
- unconditional positive thermodynamic dissipation;
- a unique globally weakest named axiom list;
- novelty or priority.

These are scope boundaries of the conditional results. The finite model in the Mathematical Supplement, Part III, has an assignment-neutral genesis stage followed by a post-genesis causal-certificate construction. Global post-genesis equivariance is not asserted. Its support-loop effect and actual future-capacity effect are certified against different events.

The correspondence methodology inherits these boundaries. It does not establish that the mathematical axioms uniquely describe nature. Literature classifications can change with new protocol information; a systematic negative literature claim would require an independently specified search universe and screening record. The RNA-routing design remains prospective.

17.2 Related work

The mathematical ingredients used here have substantial established precedents. Structural equations and surgical intervention semantics belong to the structural-causal-model tradition [1]. The total-variation contraction arguments use classical Markov-kernel / coefficient-of-ergodicity methods [2]. Equivariant symmetry and symmetry breaking have a large mathematical literature [3].

For the resource section, ordinary **viability kernels** are established in control/set-valued dynamics [4,5]. More importantly, the CRK quantifier

$$\forall \text{ admissible move/disturbance } \exists \text{ causal response remaining viable}$$

is structurally close to the established **discriminating kernel / maximal robust control invariant set** of robust viability theory and dynamic games [6,7]. The present manuscript therefore does not claim to invent viability, robust controlled invariance, or discriminating-kernel reasoning. Any novelty claim for CRK would require a specialist comparison of the exact commutative-monoid residual game and equivalence theorem against that literature.

The resemblance to robust viability concerns the ordering of current disturbances and admissible responses and the requirement of remaining in an invariant set. In the residual game considered here, the current move is available when the response is selected, but future moves are not. This information pattern must be distinguished from one fixed control required to work for every disturbance. The exact statements in Sections 10–11 concern the declared commutative-monoid residual equations; no equivalence to all formulations of robust control is asserted.

The finite-sample concentration bound is an elementary Hoeffding/union-bound construction [8], and ordered/residuated algebra has an extensive independent literature [9].

There is likewise substantial prior work on coding and semantic self-organization. Assignment-catalyst models date at least to Badian [10]; Wills studied self-organization of genetic coding [11]; Füchslin and McCaskill gave an evolutionary cell-free coding model [12]; Görlich and Dittrich formalized molecular codes in reaction networks [13]; and Clark, Hickinbotham, and Stepney demonstrated semantic closure in an artificial chemistry [14].

The word and concept **reflexivity** also have direct prior territory. Carter and Wills explicitly discuss interdependence and reflexivity in the evolution of genetic coding [15], and Wills’ later review places reflexive/self-guided molecular organization in a broader origin-of-coding context [16].

Görlich and Dittrich [13] define molecular codes through contingent mappings and alternative molecular contexts in reaction networks. The assay used here additionally specifies organization interventions and challenged response laws. Structural reaction-network codability is not identified with a demonstrated origin history, causal heredity, or output-mediated maintenance.

Beckers and Halpern [29] formalize causal-model abstraction through compatible state and intervention mappings, including transformations closed under composition. The representation invariances used here concern transport of explicitly fixed scientific data. Our assembly problem instead concerns locally certified causal blocks and their specified shared boundaries. We do not identify that problem with abstraction between levels or claim to introduce intervention-preserving causal representation.

A more direct antecedent for joint causal compatibility is the structural causal marginal problem of Gresele et al. [32]. They study joint categorical structural causal models that are counterfactually consistent with marginal models over overlapping sets of variables. The present construction fixes local certificates, required and combined intervention regimes, boundary continuations, and re-entry conditions. It supplies conditions for this specified assembly problem rather than claiming that causal-model merging is a new problem or that the present result subsumes the earlier counterfactual-consistency framework.

Kolchinsky and Wolpert [30] quantify semantic information through interventions on system–environment correlations and their consequences for self-maintenance. Their framework also allows viability to be measured by membership probability in a specified viability set. Here, operational assignment, origin provenance, and assignment-specific heredity remain separate requirements, while the maintenance comparison is tied to a declared output-to-support intervention and an actual future assay. No general equivalence

between the two frameworks is asserted.

Compatible local probability laws need not admit a common global distribution. Abramsky and Brandenburger [31] give a general treatment in terms of global sections and contextuality. The probability obstruction in Mathematical Supplement S6.3 applies this established local-to-global distinction to declared overlap coordinates, with an independent-context extension preserving the local certificates. The generic obstruction is not presented as a new probability principle.

Nunes Palmeira et al. [34] model the emergence of functional coding sequences and heredity through growth selection in protocells with specified translation and catalytic processes. This addresses sequence evolution under supplied molecular-process rules. The present framework instead separates operational assignment, origin provenance and hereditary transmission as distinct certification obligations. The existence of growth selection in that model is not, by itself, evidence of a target-mapping reward, and no complete correspondence between its assumptions and the present certificates is asserted.

The model-specific construction in [28] is the direct scientific predecessor of the present article. It supplies one explicit, independently assignment-symmetric reaction-compartment witness in which contingent read-out, causal heredity, and output-mediated maintenance coexist on a common admissible parameter regime without a mapping-quality reward. The present manuscript does not present that earlier model-specific existence result as a new contribution and does not promote its particular species, reaction grammar, parameterization, or compartment mechanism to universal requirements. Its distinct contribution is the causal-interventional decomposition of the claim set, the associated identifiability and nonfactorization results, the exact composition and finite-error conditions, a separate finite non-vacuity construction within the present formalism, and the prospective experimental-certification methodology.

Accordingly, the present manuscript is positioned around its particular causal-interventional theorem architecture and proved scope: claim separation, identifiability no-gos, exact nonfactorization/composition burdens, future-capacity reflexive certification, finite-error hard/soft separation, the CRK equivalence inside the declared algebraic game, and library-relative minimality.

The individual probabilistic, symmetry, order-theoretic, and viability tools have established precedents. The contribution lies in their specified causal-interventional formulation, the explicit finite joint realization, and the prospective experimental specification. No claim of priority is made.

18. Conclusion

Contingent assignment, causal heredity, and reflexive maintenance are separate causal properties. The framework gives explicit model conditions and counterexamples rather than identifying these properties with correlation, fixed specificity, replication, or generic persistence. It separates symmetry of an ensemble from symmetry of a realization; property existence from identifiability; support-loop re-entry from actual future assignment capacity; and local certification from exact global composition. Finite-error arguments preserve structural requirements, resource claims require explicit accounting, and minimality remains relative to a declared premise language.

The experimental certification method carries these distinctions into physical assessment. A system’s membership in a hypothesis class is separate from whether the available protocol can determine that membership and from the strength of its empirical evidence. Structural violations cannot be offset by statistical precision, incompatible local assays cannot be combined after the fact, and development data cannot also supply confirmatory mapping evidence.

The resulting contribution is a mathematical and methodological account with an explicit finite model and a prospective experimental specification. The selected literature provides component precedents; the proposed RNA-routing implementation still requires measurement. Whether a real or prebiotically plausible system realizes the full hypothesis class remains open.

Author contributions, AI use, and research status

Kai Liang initiated the research, selected the questions and high-level constraints, directed continuation and critical review, and retained decisions on research scope and dissemination. The technical mathematical formalization, derivations, proofs, counterexamples, internal mathematical checking, literature analysis, and technical writing were performed using ChatGPT (OpenAI). These activities are not attributed to the human author as independently performed technical work. The author has not independently verified the mathematics and accepts responsibility for the manuscript.

Temporary ChatGPT conversations were used for context-isolated adversarial review. These were not independent AI systems, cross-model replications, human peer review, or laboratory validation. Model output alone is not evidence of mathematical correctness; the stated claims rest on their printed assumptions and arguments. This document is an unreviewed research manuscript.

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Data and materials. This article reports no new laboratory dataset. The accompanying Mathematical Supplement contains the mathematical constructions, counterexamples, and detailed conditions. The Experimental Methods Supplement contains the prospective method and unresolved implementation requirements.

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